Weak forms and problem setup — 2-D transient heat conduction on a unit square

1 Problem setup

1.1 Geometry, boundary conditions, and source

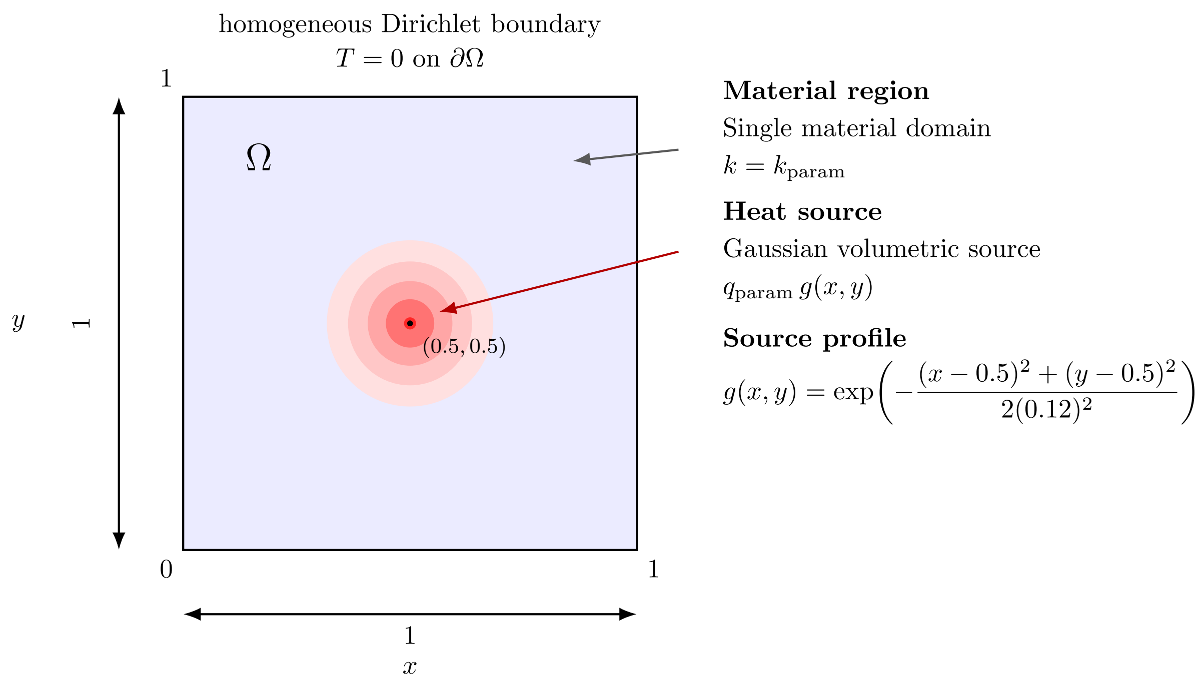

Figure 1: Unit-square domain with homogeneous Dirichlet boundary conditions and a Gaussian volumetric heat source centered at $(0.5, 0.5)$.

Let

$$\Omega = (0,1) \times (0,1) \subset \mathbb{R}^2, \qquad \Gamma = \partial\Omega.$$

The temperature field is denoted by $T(\mathbf{x}, t)$, where $\mathbf{x} = (x, y) \in \Omega$ and $t \in [0, t_{\text{end}}]$.
The code uses a single material region, so the conductivity is spatially uniform and depends only on the scalar parameter k_{param}:

$$k(\mathbf{x}; \mu) = k_{\text{param}}.$$

The volumetric source is a Gaussian centered at $(0.5, 0.5)$ with width $\sigma = 0.12$, scaled by the source parameter q_{param}:

$$f(\mathbf{x}; \mu) = q_{\text{param}}\, g(x, y),$$

with

$$g(x, y) = \exp\left(-\frac{(x-0.5)^2 + (y-0.5)^2}{2\sigma^2}\right), \qquad \sigma = 0.12.$$

The admissible parameter vector is

$$\mu = (k_{\text{param}}, q_{\text{param}}),$$

with

$$k_{\text{param}} \in [0.1, 5.0], \qquad q_{\text{param}} \in [0.0, 5.0].$$

The strong problem is

$$\frac{\partial T}{\partial t} - \nabla \cdot \big(k_{\text{param}} \nabla T\big) = q_{\text{param}}\, g(x, y) \qquad \text{in } \Omega \times (0, t_{\text{end}}], \tag{1}$$

$$T = 0 \qquad \text{on } \Gamma \times (0, t_{\text{end}}], \tag{2}$$

$$T(\mathbf{x}, 0) = 0 \qquad \text{in } \Omega. \tag{3}$$

2 Weak formulation

2.1 Function space

Define the trial and test space

$$V := H_0^1(\Omega) = \big\{ v \in H^1(\Omega) \ : \ v = 0 \text{ on } \Gamma \big\}.$$

2.2 Variational form

For each $t \in (0, t_{\text{end}}]$, the weak problem is:
Find $T(\cdot, t) \in V$ such that

$$m\left(\frac{\partial T}{\partial t}, v\right) + a(T, v; \mu) = \ell(v; \mu) \qquad \forall v \in V, \tag{4}$$

where

$$m(w, v) = \int_\Omega w\, v \,\mathrm{d}\Omega, \tag{5}$$

$$a(T, v; \mu) = k_{\text{param}} \int_\Omega \nabla T \cdot \nabla v \,\mathrm{d}\Omega, \tag{6}$$

$$\ell(v; \mu) = q_{\text{param}} \int_\Omega g(x, y)\, v \,\mathrm{d}\Omega. \tag{7}$$

Equivalently, the weak statement can be written as

$$\int_\Omega \frac{\partial T}{\partial t}\, v \,\mathrm{d}\Omega + k_{\text{param}} \int_\Omega \nabla T \cdot \nabla v \,\mathrm{d}\Omega = q_{\text{param}} \int_\Omega g(x, y)\, v \,\mathrm{d}\Omega \qquad \forall v \in H_0^1(\Omega). \tag{8}$$

2.3 Affine parameter dependence

The formulation is affine in the two parameters. Defining

$$a_0(T, v) = \int_\Omega \nabla T \cdot \nabla v \,\mathrm{d}\Omega, \qquad \ell_0(v) = \int_\Omega g(x, y)\, v \,\mathrm{d}\Omega,$$

the parameter dependence becomes

$$a(T, v; \mu) = k_{\text{param}}\, a_0(T, v), \qquad \ell(v; \mu) = q_{\text{param}}\, \ell_0(v). \tag{9}$$

2.4 Time-discrete weak form used in the code

The implementation uses implicit Euler with

$$\Delta t = 2.0 \times 10^{-2}, \qquad t_{\mathrm{end}} = 4.0 \times 10^{-1}.$$

Let $t_n = n\Delta t$ and let $T^n \in V$ denote the temperature at time level t_n. Then the fully discrete weak problem is:
Find $T^{n+1} \in V$ such that

$$\int_\Omega \frac{T^{n+1} - T^n}{\Delta t} \, v \, \mathrm{d}\Omega + k_{\mathrm{param}} \int_\Omega \nabla T^{n+1} \cdot \nabla v \, \mathrm{d}\Omega = q_{\mathrm{param}} \int_\Omega g(x, y) \, v \, \mathrm{d}\Omega \qquad \forall v \in V. \tag{10}$$

2.5 Finite-element space used in the code

The domain is discretized by a structured 25×25 partition of the unit square, with each rectangular cell split into two triangles. A conforming scalar $\mathbb{P}_1$ Lagrange basis is used for the temperature field, and all boundary degrees of freedom are fixed to the homogeneous Dirichlet value.